

DBMS WEEK 4

TOPIC: JOINS AND SET OPERATIONS

STEP 1: Create Table

```
mysql> create database task4;
Query OK, 1 row affected (0.03 sec)

mysql> use task4;
Database changed
mysql> create table class (
  -> id int,
  -> name varchar(30)
  -> );
Query OK, 0 rows affected (0.06 sec)

mysql> desc class;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| id    | int           | YES  |     | NULL    |       |
| name  | varchar(30)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.01 sec)
```

STEP 2: Insert Data

```
mysql> insert into class values
  -> (1,'abhi'),
  -> (2,'adam'),
  -> (4,'alex');
Query OK, 3 rows affected (0.03 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> select * from class;
+----+-----+
| id | name |
+----+-----+
|  1 | abhi |
|  2 | adam |
|  4 | alex |
+----+-----+
3 rows in set (0.00 sec)
```

QUESTION :1 – Cross Join

```
mysql> insert into class_info values
  -> (1,'delhi'),
  -> (2,'mumbai'),
  -> (3,'chennai');
Query OK, 3 rows affected (0.03 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> select * from class_info;
+----+-----+
| id | address |
+----+-----+
|  1 | delhi   |
|  2 | mumbai  |
|  3 | chennai |
+----+-----+
3 rows in set (0.00 sec)
```

PART 2: Inner Join

```
mysql> create table class (
-> id int,
-> name varchar(30)
-> );
Query OK, 0 rows affected (0.06 sec)
```

```
mysql> desc class;
```

Field	Type	Null	Key	Default	Extra
id	int	YES		NULL	
name	varchar(30)	YES		NULL	

2 rows in set (0.01 sec)

```
mysql> create table class_info (
-> id int,
-> address varchar(30)
-> );
Query OK, 0 rows affected (0.06 sec)
```

```
mysql> desc class_info;
```

Field	Type	Null	Key	Default	Extra
id	int	YES		NULL	
address	varchar(30)	YES		NULL	

2 rows in set (0.00 sec)

```
mysql> insert into class_info values
-> (1,'delhi'),
-> (2,'mumbai'),
-> (3,'chennai');
Query OK, 3 rows affected (0.03 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

```
mysql> select * from class_info;
```

id	address
1	delhi
2	mumbai
3	chennai

3 rows in set (0.00 sec)

```
mysql> insert into class values
-> (1,'abhi'),
-> (2,'adam'),
-> (3,'alex'),
-> (4,'anu');
Query OK, 4 rows affected (0.03 sec)
Records: 4 Duplicates: 0 Warnings: 0
```

```
mysql> select * from class;
```

id	name
1	abhi
2	adam
3	alex
4	anu

4 rows in set (0.00 sec)

QUESTION :2 – Inner Join

```
mysql> select * from class
-> inner join class_info
-> on class.id = class_info.id;
+-----+-----+-----+-----+
| id | name | id | address |
+-----+-----+-----+-----+
| 1 | abhi | 1 | delhi |
| 2 | adam | 2 | mumbai |
| 3 | alex | 3 | chennai |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

QUESTION :3 – Inner Join

```
mysql> select class.name,
-> class_info.address
-> from class inner join class_info
-> on class.id = class_info.id;
+-----+-----+
| name | address |
+-----+-----+
| abhi | delhi |
| adam | mumbai |
| alex | chennai |
+-----+-----+
3 rows in set (0.00 sec)
```

PART 3: Natural Join

QUESTION: 4 - Natural Join

```
mysql> select*from class
-> natural join class_info;
+-----+-----+-----+
| id | name | address |
+-----+-----+-----+
| 1 | abhi | delhi |
| 2 | adam | mumbai |
| 3 | alex | chennai |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

PART 4: Left Outer Join

```
mysql> select * from class;
+-----+-----+
| id | name |
+-----+-----+
| 1 | abhi |
| 2 | adam |
| 3 | alex |
| 4 | anu |
| 5 | ashish |
+-----+-----+
5 rows in set (0.00 sec)

mysql> select * from class_info;
+-----+-----+
| id | address |
+-----+-----+
| 1 | delhi |
| 2 | mumbai |
| 3 | chennai |
| 7 | noida |
| 8 | panipat |
+-----+-----+
5 rows in set (0.00 sec)
```

QUESTION :5 – Left Join

```
mysql> select*from class
-> left outer join class_info
-> on class.id = class_info.id;
```

id	name	id	address
1	abhi	1	delhi
2	adam	2	mumbai
3	alex	3	chennai
4	anu	NULL	NULL
5	ashish	NULL	NULL

5 rows in set (0.02 sec)

QUESTION :6 – Left Join

```
mysql> select*from class
-> left join class_info
-> on class.id = class_info.id;
-> where class_info.id is null;
```

id	name	id	address
4	anu	NULL	NULL
5	ashish	NULL	NULL

2 rows in set (0.00 sec)

PART 5: Right Join

QUESTION :7 – Right Join

```
mysql> select*from class
-> right outer join class_info
-> on class.id = class_info.id;
```

id	name	id	address
1	abhi	1	delhi
2	adam	2	mumbai
3	alex	3	chennai
NULL	NULL	7	noida
NULL	NULL	8	panipat

5 rows in set (0.00 sec)

QUESTION :8 – Right Join

```
mysql> select*from class
-> right join class_info
-> on class.id = class_info.id
-> where class.id is null;
```

id	name	id	address
NULL	NULL	7	noida
NULL	NULL	8	panipat

2 rows in set (0.00 sec)

PART 6: Full Outer Join

QUESTION :9 - Full Outer Join

```
mysql> select*from class
-> left join class_info
-> on class.id = class_info.id;
```

id	name	id	address
1	abhi	1	delhi
2	adam	2	mumbai
3	alex	3	chennai
4	anu	NULL	NULL
5	ashish	NULL	NULL

5 rows in set (0.00 sec)

QUESTION :10 - Full Outer Join

```
mysql> select*from class
-> left join class_info
-> on class.id = class_info.id
-> where class_info.id is null
-> union
-> select*from class
-> right join class_info
-> on class.id = class_info.id
-> where class.id is null;
```

id	name	id	address
4	anu	NULL	NULL
5	ashish	NULL	NULL
NULL	NULL	7	noida
NULL	NULL	8	panipat

4 rows in set (0.03 sec)

PART 7: Union

```
mysql> select * from first_table;
```

id	ame
1	abhi
2	adam

2 rows in set (0.00 sec)

```
mysql> select * from second_table;
```

id	ame
2	adam
3	chester

2 rows in set (0.02 sec)

QUESTION :11 – Union

```
mysql> select * from first_table
-> union
-> select * from second_table;
```

id	ame
1	abhi
2	adam
3	chester

3 rows in set (0.00 sec)

QUESTION :12 – Union

```
mysql> select ame from first_table
-> union
-> select ame from second_table;
```

ame
abhi
adam
chester

3 rows in set (0.00 sec)

PART 8: Union All

QUESTION :13 - Union All

```
mysql> select * from first_table
-> union all
-> select * from second_table;
```

id	ame
1	abhi
2	adam
2	adam
3	chester

4 rows in set (0.00 sec)

QUESTION :14 - Union All

```
mysql> select count(*)
-> from
-> (
-> select * from first_table
-> union all
-> select * from second_table
-> ) a;
```

count(*)
4

1 row in set (0.02 sec)

PART 9: Intersect

QUESTION :15 – Intersect

```
mysql> select f.*
-> from first_table f
-> inner join second_table s
-> on f.id = s.id
-> and f.ame = s.ame;
+-----+
| id | ame |
+-----+
| 2 | adam |
+-----+
1 row in set (0.02 sec)
```

QUESTION :16 – Intersect

```
mysql> select f.ame
-> from first_table f
-> inner join second_table s
-> on f.ame = s.ame;
+-----+
| ame |
+-----+
| adam |
+-----+
1 row in set (0.00 sec)
```

PART 10: Minus

QUESTION :17 – Minus

```
mysql> select f.*
-> from first_table f
-> left join second_table s
-> on f.id = s.id
-> and f.ame = s.ame
-> where s.id is null;
+-----+
| id | ame |
+-----+
| 1 | abhi |
+-----+
1 row in set (0.00 sec)
```

QUESTION :18 – Minus

```
mysql> select f.ame
      -> from first_table f
      -> left join second_table s
      -> on f.ame = s.ame
      -> where s.id is null;
```

ame
abhi

1 row in set (0.00 sec)

ADVANCED INTERVIEW QUESTIONS

QUESTION 19

```
mysql> select c.id, c.name, ci.address
      -> from class c
      -> inner join class_info ci
      -> on c.id = ci.id;
```

id	name	address
1	abhi	delhi
2	adam	mumbai
3	alex	chennai

3 rows in set (0.00 sec)

QUESTION 20

```
mysql> select c.id, c.name, case
      -> when ci.address is null
      -> then 'address missing'
      -> else 'address available'
      -> end as status
      -> from class c
      -> left join class_info ci
      -> on c.id = ci.id;
```

id	name	status
1	abhi	address available
2	adam	address available
3	alex	address available
4	anu	address missing
5	ashish	address missing

5 rows in set (0.02 sec)